

Joseph Walton

Senior AI Engineer | Generative AI | Agentic AI | Azure

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GitHub: github.com/jwgwalton Professional Summary

Senior AI Engineer with 10+ years' experience delivering production AI solutions across regulated government, financial services, intellectual property and defence environments. Hands-on expertise in Generative AI, RAG, agentic AI, Azure OpenAI, Microsoft AI Foundry, LangChain and LangGraph, with strong Python and FastAPI development experience. Experienced in prompt engineering, LLM evaluation and observability, MLOps, secure deployment and enterprise integration. Proven technical leader with production solutions delivering £2M+ annual savings and significant improvements in performance and cost. Technical Skills

Generative AI: Azure OpenAI, Microsoft AI Foundry, LangChain, LangGraph, RAG, agentic AI  
Engineering: Python, FastAPI, REST APIs, SQL, Java, microservices, Azure Functions  
Azure & Data: Azure AI Search, Cognitive Services, Azure Databricks, Apache Spark, MLflow  
MLOps: Azure ML Pipelines, Vertex AI Pipelines, MLflow, Docker, CI/CD, model evaluation  
Security: Secure by Design, AI red teaming, adversarial testing, secure containerisation  
Machine Learning: Hugging Face, TensorFlow, PyTorch, Scikit-learn

Professional Experience IBM

Lead Data Scientist June 2025 – Present | Contract – Central Government

Designed and delivered a production agentic AI solution using Microsoft AI Foundry  
Built an MCP server exposing a data-matching algorithm for reuse across government  
Developed prompt engineering, LLM evaluation and observability practices to improve  
Built human-in-the-loop LLM evaluation workflows, reducing evaluation effort by 50%  
Re-engineered a data-matching algorithm using Polars, reducing runtime from two hours to 10 minutes  
Worked with architecture, engineering and business stakeholders to deliver secure, scalable solutions

Tech Stack: Python, Microsoft AI Foundry, Azure OpenAI, LangGraph, MCP, Azure Functions, Polars  
Intellectual Property Office

Senior Machine Learning Engineer November 2021 – April 2025

Developed production Generative AI applications using Python, LangChain, Azure OpenAI  
Built a RAG agent for identifying relevant Goods and Services terms, integrating LLMs  
Implemented and optimised prompt engineering and LLM evaluation, improving application performance  
Tuned Elasticsearch HNSW vector search, improving search and recommendation relevance  
Implemented end-to-end MLOps and CI/CD across multiple environments, reducing model deployment time  
Conducted Generative AI red-team exercises and advised technical and non-technical stakeholders

Tech Stack: Python, Azure OpenAI, LangChain, Azure Functions, Azure AI Search, Azure ML Pipelines,  
Elasticsearch, Azure Databricks Earlier Experience Incopro

Lead Machine Learning Engineer / Machine Learning Engineer November 2018 – November 2021

Led delivery of secure production ML systems using Python, FastAPI, Hugging Face, TensorFlow  
Developed and deployed scalable Python/FastAPI services supporting machine learning operations  
Designed an end-to-end MLOps framework using Vertex AI Pipelines, orchestrating training and deployment  
Developed multilingual NLP models covering 90+ languages and built scalable text classification systems  
Mentored engineers transitioning into machine learning roles.

Tech Stack: Python, FastAPI, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, MLflow, Docker, GCP  
Office for National Statistics

Machine Learning Engineer September 2017 – November 2018

Developed Python ML services and scalable data pipelines processing sensitive administrative data.  
Contributed to the backend architecture for the UK Census 2021 data-collection system.

Barclays

Data Scientist — Secondment September 2018 – October 2018

Applied statistical and analytical methods to securely use payments data for regional analysis.  
Member of the winning ONS × Barclays Data Hackathon team.

Purple Secure Systems

Software Engineer August 2014 – October 2016

Developed Python and Java big-data processing systems for government and defence contractors.

Education University of Bath

MSc Applied Mathematics — 2016–2017 BSc Physics — 2010–2013 Award

Winner — ONS × Barclays Data Hackathon Used payments data to develop more granular and timely estimates of Gross Value Added (GVA).